Trainings

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

When the engine is operating with the rocker lever covers removed, oil splashing can cause personal injury. To reduce the possiblity of personal injury, wear eye and face protection.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)

- Remove the water connections for the wet exhaust manifolds. Refer to Procedure 011-016 in Section 11. (/qs3/pubsys2/xml/en/procedures/28/28-011-016-tr.html)
- Remove the air crossover. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html)
- Remove the outboard aftercoolers. Refer to Procedure 010-004 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-004-tr.html)
- Remove the step timing control (STC) oil manifold. Refer to Procedure 006-038 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-038-tr.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
- Remove the crossheads. Refer to Procedure 002-001 in Section 2. (/qs3/pubsys2/xml/en/procedures/28/28-002-001-tr.html)
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html)
- Remove the push rods or tubes. Refer to Procedure 004-014 in Section 14. (/qs3/pubsys2/xml/en/procedures/28/28-004-014-tr.html)

with Electronically Actuated Injector

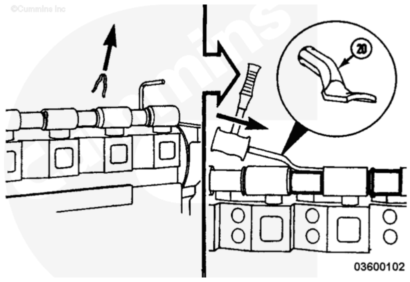
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Remove the injector supply lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html)
- Remove the rocker lever covers. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html)
- Remove the push rods or tubes. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/28/28-004-014-tr.html)
- Remove the crossheads. Refer to Procedure 002-001 in Section 2. (/qs3/pubsys2/xml/en/procedures/28/28-002-001-tr.html)
- Remove the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)
- Remove any aftercooler assembly plumbing that obstructs the removal of the rocker lever housing. Refer to Procedure 010-002, (/qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html) Refer to Procedure 010-003. and (/qs3/pubsys2/xml/en/procedures/28/28-010-003-tr.html) Refer to Procedure 010-004 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-004-tr.html)

Remove

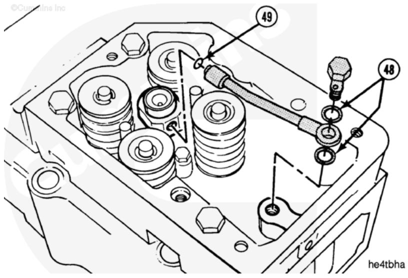
with Mechanically Actuated Injector

Note : Older engines may contain clips on the water tubes. Remove the clips.

Use the water tube driver, Part Number ST-1319 (20). Drive the tubes on the right bank toward the rear of the engine. Drive the tubes on the left bank toward the front of the engine. Drive the tube until it clears the rocker lever housing.

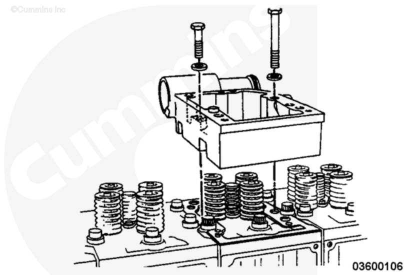


Remove the STC banjo connector mounting screw. Remove the banjo connector. Remove and discard the sealing washers (48) and the o-ring (49).



Remove the six mounting capscrews and the rocker lever housing.

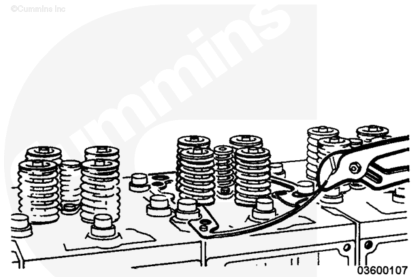
Note : Note the placement of the single shorter capscrew.



⚠ CAUTION ⚠

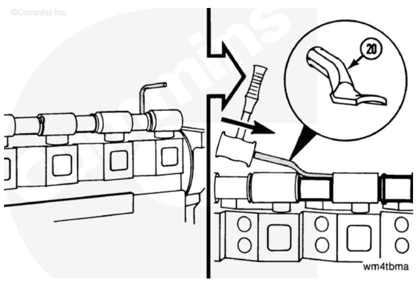
Some rocker lever housing gaskets have a thin metal core. Handle with care to reduce the possibility of personal injury.

Use pliers. Remove and discard the gasket.



with Electronically Actuated Injector

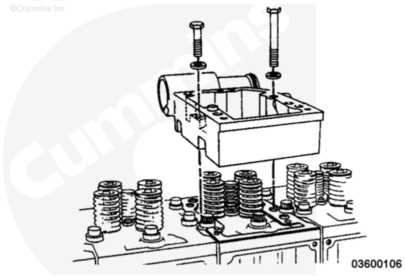
Use the Water Tube Driver, Part Number ST-1319. Drive the tubes on the right bank toward the rear of the engine. Drive the tubes on the left bank toward the front of the engine. Drive the tube until it clears the rocker lever housing.



Remove the six mounting capscrews, plain washers, and the rocker lever housing.

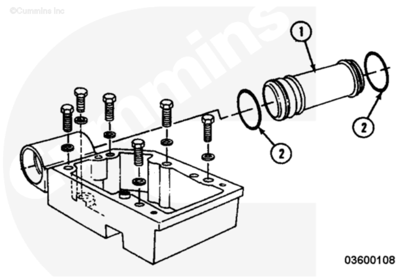
Note : Note the placement of the single shorter capscrew.

All rocker lever housings are **not** the same. Tag or mark the parts with their respective locations relative to the cylinder numbers to aid during assembly.

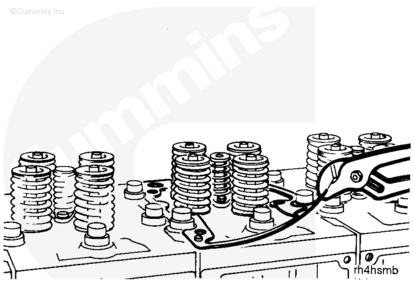


Remove the water transfer tube (1) from the rocker lever housing and discard the o-rings (2).

All water transfer tubes are **not** the same. Tag or mark the parts with their respective locations relative to the cylinder numbers to aid during assembly.



Remove and discard the rocker lever housing gasket.



Clean and Inspect for Reuse

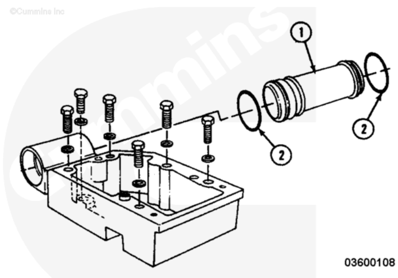
with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

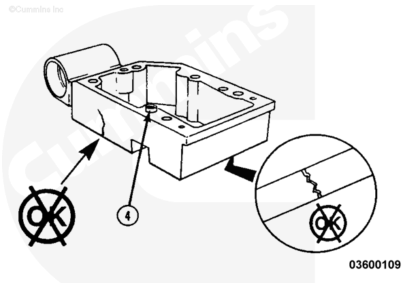
Remove the parts.

- Water transfer tube
- O-rings - discard the o-rings.

Use steam or solvent to clean the parts.

Check the parts for damage. If the ring dowel (4) is damaged, it **must** be replaced. Use the blind hole puller, contained in Light Duty Puller Kit, Part Number 3375784 or equivalent, to remove the dowel ring.

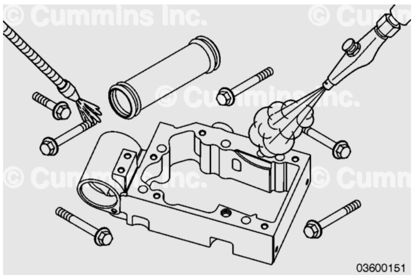
Use the dye penetrant method and check the housing for cracks. Discard the housing if it is cracked.



with Electronically Actuated Injector

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

⚠ WARNING ⚠

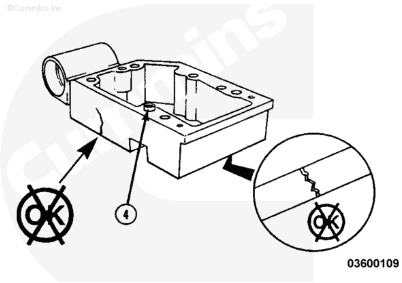
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam or solvent to clean the parts.

Dry with compressed air.

Clean the rocker lever housing mating surface on the engine with a nylon abrasive pad, Scotch-Brite™ 7448, or equivalent.

Inspect the parts for damage. If the ring dowel (4) is damaged, it **must** be replaced. Use a blind hole puller contained in Light Duty Puller Kit, Part Number 3375784, or equivalent, to remove the dowel ring.

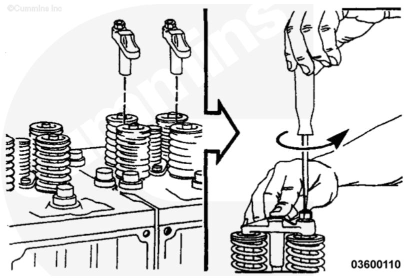


Use the Crack Detection Kit, Part Number 3375432, to inspect the housing for cracks. If the housing is cracked, it **must** be replaced.

Install

with Mechanically Actuated Injector

Install the crossheads.



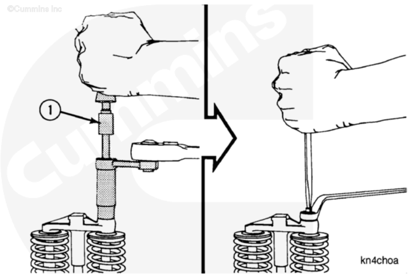
Lubricate the crosshead guides and valve stems with engine oil.

Note : The adjusting screws must be toward the exhaust side of the cylinder head.

Install and adjust the crossheads. Push down on the crosshead and slowly turn the adjusting screw until it touches the valve stem. Do **not** turn far enough to raise the crosshead.

⚠ CAUTION ⚠

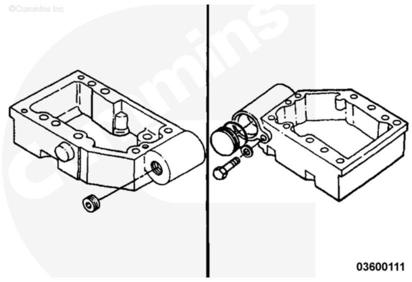
The adjusting screw must not turn when the locknut is tightened.



Tighten the locknut to the value shown. The torque values given are with and without the use of torque wrench adapter, Part Number ST-669.

		N•m	ft-lb
Torque Value:	With adapter	35	26
Torque Value:	Without adapter	40	30

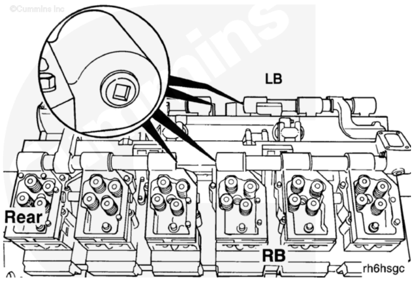
Note : Some engines will contain special rocker lever housings and water transfer tubes. See the following information.



Some engines contain rocker lever housings with threaded holes for clamps and support brackets for the application. Refer to any notes or photographs made during engine disassembly for the location of the special housings.

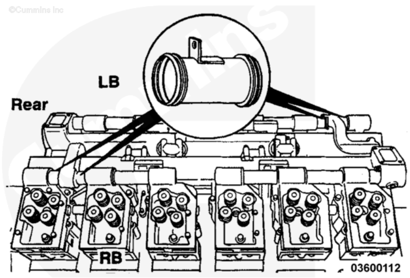
The housings that are installed closest to the rear of the engine will contain holes for pipe plugs or aftercooler water tube fittings. Older housings used at the rear of the engine will contain a plug with an o-ring seal and a tab that is fastened to the housing with a capscrew.

If the engine contains a wet exhaust manifold, the housings with a pipe plug instead of the bore for the water tube **must** be installed in the center of the engine so that the engine coolant will flow through the exhaust manifold.



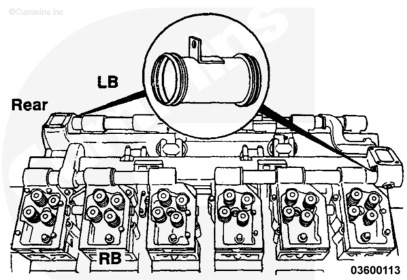
On K38 engines the housings with the plugs **must** be installed on cylinder number 3 and 4 of both banks. On K50 engines, the housings with the plugs **must** be installed on cylinder number 4, and cylinder number 5 of both banks.

If the engine contains a wet exhaust manifold, the water tubes installed between the housings for the front two cylinders on the left bank and the rear two cylinders on the right bank are different.



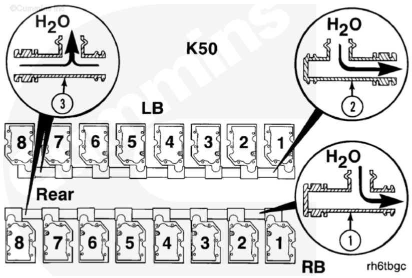
These water tubes will be shorter than the other tubes and contain a tab with a capscrew hole to allow the installation of the water connection for the exhaust manifold.

The water tube with a tab for a capscrew **must** also be installed in the housing for the right bank front and left bank rear cylinders. The capscrew will hold this tube in a bore in the wet exhaust manifold.



Some engines have special water tubes that allow the engine coolant to flow through a torque converter cooler.

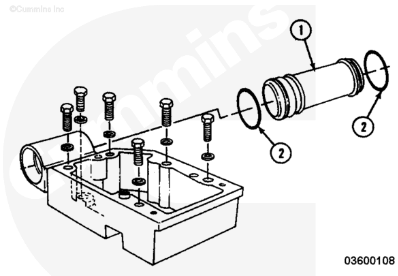
Two of the tubes contain plugs. The tubes with the plugs are different for the right bank (1) and the left bank (2). These tubes **must** be installed between the Number 1 and Number 2 cylinders so that the end with the plug is in the bore of the housing for the Number 2 cylinder, (both toward the rear of the engine). The tubes (3) that are installed between the housings for the two cylinders nearest to the rear of the engine on both banks **must not** contain the plug. The tubes nearest to the rear are the same for both banks.



Lubricate the o-ring seals (2) with vegetable oil. Install the o-rings.

Install the water transfer tube (1).

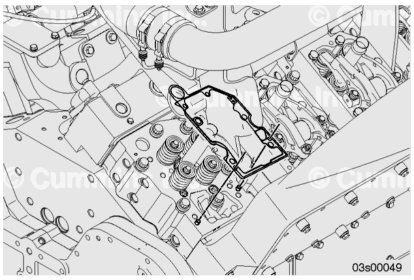
Note : For the left bank housing, the ends of the tubes with two grooves must be installed in the bore that is positioned toward the front of the engine. For the right bank housing, the ends of the tubes with two grooves must be installed in the bore that is positioned toward the rear of the engine.



Note : The gasket must be aligned correctly with the dowel pins.

Make sure the cylinder head top face is free from debris and grease.

Install the gasket to the cylinder head.



Note : The housing must be aligned correctly with the dowel pins

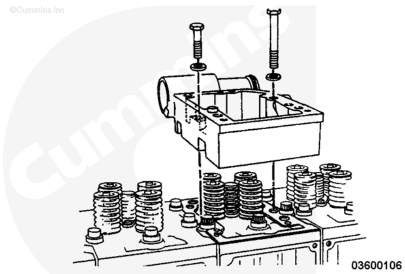
Make sure the rocker housing bottom face is free from debris and grease.

Install the housing.

Check the threads of the rocker housing capscrews for damage. Check under the capscrew heads for cracks. Check for any deformation or necking of the capscrew.

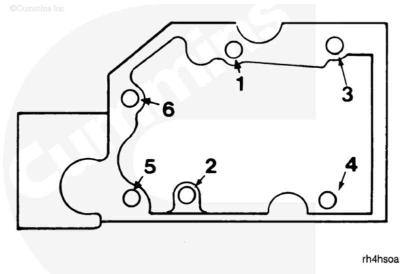
Note : One of the capscrews is shorter than the other five.

Install the six capscrews.



Use the sequence shown to tighten the capscrews. Tighten the capscrews.

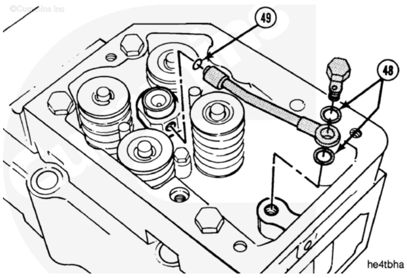
Torque Value: 122 n•m [90 ft-lb]



Lubricate the o-ring with engine oil. Install the o-ring (49) on the jumper tube.

Install the jumper tube, copper sealing washers (48), and connector screw. Tighten the screw.

Torque Value: 25 n•m [221 in-lb]



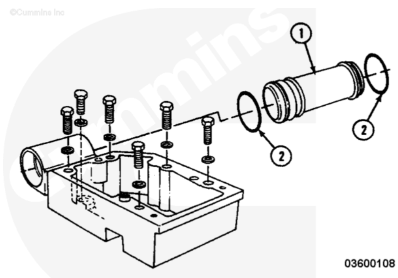
with Electronically Actuated Injector

Use vegetable oil to lubricate the o-rings (2).

Use the minimum amount of stretch and check that the o-rings do **not** twist or kink. Install the o-rings (2) into the grooves on the outside diameter at each end of the water transfer tubes (1).

Note : Check that the tagged or labeled water transfer tubes are installed in the correct locations.

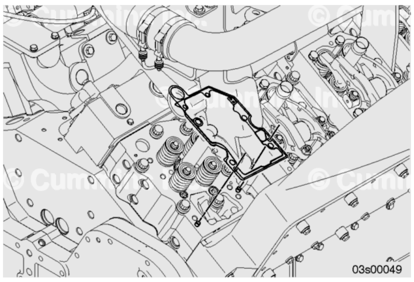
Install the water transfer tube (1).



Note : The gasket must be aligned correctly with the dowel pins.

Make sure the cylinder head top face is free from debris and grease.

Install the gasket to the cylinder head.

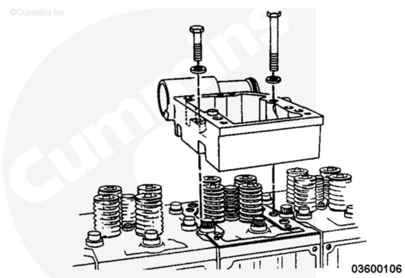


Note : Check that the tagged or labeled rocker lever housings are installed in the correct locations.

Note : The housing must be aligned correctly with the dowel pins.

Make sure the rocker housing bottom face is free from debris and grease`.

Install the housing.



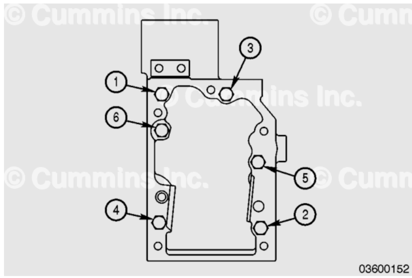
Note : One of the capscrews is shorter than the other five.

Check the threads of the rocker housing capscrews for damage. Check under the capscrew heads for cracks. Check for any deformation or necking of the capscrew.

Install the six capscrews.

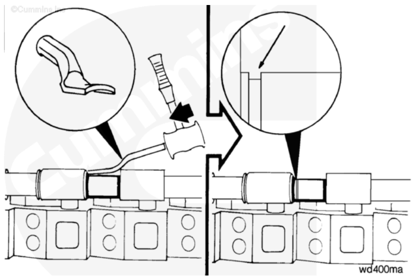
Use the sequence shown to tighten the capscrews.

Torque Value: 122 n•m [90 ft-lb]



⚠ CAUTION ⚠

Check that the o-rings are not cut, damaged, or rolled over when driving the water transfer tubes into the adjacent rocker lever housings.



Use water tube driver, Part Number ST-1319, and a hammer to drive the water transfer tube into the adjacent rocker lever housing until the water transfer tube is approximately centered.

Finishing Steps

with Mechanically Actuated Injector

- Install the water connections for the wet exhaust manifolds. Refer to Procedure 011-016 in Section 11. (/qs3/pubsys2/xml/en/procedures/28/28-011-016-tr.html)
- Install the push rods or tubes. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/28/28-004-014-tr.html)
- Install the rocker lever assembly. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html)
- Adjust the overhead. Refer to Procedure 003-006 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)

- Install the STC oil manifold. Refer to Procedure 006-038 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-038-tr.html)
- Install the outboard aftercoolers. Refer to Procedure 010-004 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-004-tr.html)
- Install the air crossover. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Operate the engine. Check for leaks.

with Electronically Actuated Injector

- Install any aftercooler assembly plumbing that obstructs the removal of the rocker lever housing. Refer to Procedure 010-002, or (/qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html) Refer to Procedure 010-003. or (/qs3/pubsys2/xml/en/procedures/28/28-010-003-tr.html) Refer to Procedure 010-004 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-004-tr.html)
- Install the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)
- Install the push rods or tubes. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/28/28-004-014-tr.html)
- Install the crossheads. Refer to Procedure 002-001 in Section 2. (/qs3/pubsys2/xml/en/procedures/28/28-002-001-tr.html)
- Install the rocker lever assembly. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html)
- Adjust the overhead. Refer to Procedure 003-006 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html)
- Install the rocker lever covers. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
- Install the injector supply lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Operate the engine to 71°C [160°F] minimum cooling temperature. Check for leaks.

Last Modified: 27-Apr-2021